# TECHNICAL EVALUATION REPORT

## Convex Perturbation and the Suppression of the Indivisibility Deficit

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# 1 Summary of the submission

Let $\{\Gamma_0(t \leftarrow s)\}$ be a continuous two-time column-stochastic family on a finite configuration space $\Omega = \{1, \ldots, N\}$, and let $\{\Gamma_{\mathrm{ref}}\}$ be *divisible*. The paper studies the convex perturbation

$$\Gamma_g(t \leftarrow s) \;=\; (1-g)\,\Gamma_0(t \leftarrow s) \;+\; g\,\Gamma_{\mathrm{ref}}(t \leftarrow s), \qquad g \in [0,1], \tag{1}$$

and asks how the Chapman–Kolmogorov (CK) deficit $\Delta(t,s,r;\Gamma_g) = \Gamma_g(t \leftarrow s) - \Gamma_g(t \leftarrow r)\Gamma_g(r \leftarrow s)$ and its operator norm $D(t,s,r;\Gamma_g)$ respond to the coupling $g$. Detectability is indexed by temporal resolution through $D_{\mathrm{CK}}(\delta t;\Gamma) = \sup\{D(t,s,r;\Gamma) : s < r < t,\ t - s \le \delta t\}$.

Four contributions are claimed in §1: the convex perturbation as *the canonical model* of a dynamically consistent, state-conditioned interaction; an exact algebraic identity for the perturbed deficit; a two-threshold formulation of empirical detectability forced by non-monotonicity; and an interval-local stability result for the first-crossing threshold. Sections §5 and §6 add a causal lift with quadratic suppression of an integrated deficit, and extensions to state-dependent and nonlinear coupling.

This is a well-posed perturbation problem with a clean answer, and the paper is to be judged on three separate axes: whether the mathematics is correct, whether the perturbation model is what the paper says it is, and whether the originality claim survives contact with the existing literature on mixing away memory effects.

# 2 Overall assessment

The mathematics seems to be correct. This report independently verified the organizing identity over three hundred random instances and reproduced every reported quantity of the worked two-state example exactly, including the non-monotone rebound; the numerical verification record appears below. The subsequent operator-norm suppression bound, the Lipschitz dependence on $g$, the terminal-threshold estimate, the interval-local stability bound, the integrated-deficit estimate and the finite-dimensional state-dependent bound are all sound under their stated hypotheses. No defect identified here seems to invalidate any central algebraic or norm estimate.

The core identity is elementary (its proof is two lines of bilinear expansion) but it earns its place. Its value is that it exposes two terms with distinct structural roles: a residual substrate deficit proportional to $(1-g)$, and a substrate–reference interaction term proportional to $g(1-g)$. Because the identity is exact rather than an estimate, the possible non-monotonicity of the response becomes an algebraic fact about the family rather than an artefact of the bounding procedure, and that in turn forces the separation of a first-crossing threshold from a terminal suppression threshold. That chain of reasoning is the paper's genuine methodological contribution, and it is correctly argued (Finding 1).

Three qualifications must be pointed out, and two of them are material.

First, the originality claim is narrower than the paper's positioning implies. Convex mixing of Markovian and non-Markovian evolutions, including non-monotone changes in Markovianity, has an established literature. More consequentially, prior work already defines robustness measures through the *minimum amount of Markovian dynamics that must be mixed with a non-Markovian evolution to render it Markovian*, one of these at a constant, time-independent weight and requiring Markovianity at all times. That is conceptually very close to the manuscript's terminal-suppression question, and neither reference is cited (Finding 5). What survives, and is defensible, is narrower: the particular CK-deficit factorization for continuous two-time column-stochastic families, the

fixed-resolution $\varepsilon$-threshold formalism, the explicit sign-blind bound in terms of $D_{\mathrm{CK}}$ and $M_*$, and the state-dependent finite-dimensional extension.

Second, the physical interpretation of the perturbation is not established. Equation (1) always defines a valid two-time column-stochastic family, but the paper does not prove that this pairwise family is the transition family of a single stochastic process, nor that any measurement, control or state-feedback intervention produces this convex mixture with a *time-independent* weight $g$ (Finding 3). The mathematical perturbation theory is valid as stated; the claim that it is *the canonical model* of a dynamically consistent interaction requires qualification or an explicit realization theorem.

Third, the mathematical depth is modest: an exact two-line expansion, its triangle inequality, a linear envelope, and a compactness argument. This is not a criticism of correctness, and the paper is unusually disciplined about not overselling (§7.3, Remark 4.5 and Remark 4.10 each decline to claim more than has been established) but it bears on how much weight the work can carry.

One further item is recorded separately because it is a plain error rather than a matter of emphasis: the inference that exact restoration of divisibility occurs "that is, at $g = 1$" is not justified, and is contradicted by the paper's own worked example (Finding 2).

# 3 Section-by-section analysis

## 3.1 Framework (§2)

Definition 2.1 gives a coherent finite-dimensional setting. Lemma 2.3 correctly extends the deficit to the compact closure of the strict triple set, where the boundary condition forces $\Delta = 0$ on the three degenerate faces $r = s$, $r = t$, $t = s$; the supremum over strict triples is therefore finite and attained as a maximum. The argument is clean and the attainment is used later without further comment, correctly.

Definition 2.5 makes divisibility of the reference an *explicit structural hypothesis* rather than an inference, and §2.2 is careful on a point a hasty treatment would elide: state conditioning alone does not produce divisibility, since a collection of kernels each conditioned on the present state need not compose across intermediate times. Dynamical consistency is the additional requirement, and the paper imposes it rather than deriving it. Remark 2.8 distinguishes the construction from projection-operator and hidden-Markov augmentations: the perturbation operates on $\Omega$ itself and does not enlarge the configuration space, so the deficit of the perturbed family remains a meaningful object of study. Both points are correct to my judgment.

## 3.2 The exact deficit identity (§3)

With $A = \Gamma_0(t \leftarrow s)$, $C = \Gamma_0(t \leftarrow r)$, $E = \Gamma_0(r \leftarrow s)$ from the substrate and $D = \Gamma_{\mathrm{ref}}(t \leftarrow r)$, $F = \Gamma_{\mathrm{ref}}(r \leftarrow s)$ from the divisible reference (so that the reference long kernel is $DF$) Theorem 3.1 states

$$\Delta(t, s, r; \Gamma_g) \;=\; (1-g)\big(A - CE\big) \;-\; g(1-g)\,(C-D)(F-E). \tag{2}$$

The proof is a direct expansion turning on the coefficient collapse $(1-g)^2 + g(1-g) = (1-g)$. It is correct; its proof is elementary bilinear algebra. The triangle inequality then gives the suppression bound

$$D(t, s, r; \Gamma_g) \;\le\; (1-g)\, D(t, s, r; \Gamma_0) \;+\; g(1-g)\, M(t, s, r), \qquad M := \big\|(C-D)(F-E)\big\|_{\mathrm{op}}. \tag{3}$$

**Finding 1** (The paper's genuine methodological point)**.** Because (2) is an *exact identity* and not an estimate, failure of the response to be monotone in $g$ is an *algebraic fact* about the family rather than an artefact of the bounding procedure. Differentiating the right-hand side of (3) gives $-D(t,s,r;\Gamma_0) + (1-2g)M(t,s,r)$, non-positive throughout $[0,1]$ if and only if $D(t,s,r;\Gamma_0) \geq M(t,s,r)$. When the interaction term dominates, the bound rises before it falls; and the *exact* deficit may vanish in the interior of the coupling range and rebound.

This is what forces the two-threshold theory. A first crossing of a detection level can be followed by re-entry above it, so only a *terminal* threshold supports the operational reading "above this coupling, indivisibility is undetectable at sensitivity $\varepsilon$". The reasoning is correct, and it is the substantive methodological contribution of the paper. The qualitative phenomenon of non-monotone behaviour under convex mixing has substantial prior art in the quantum setting; what is the author's own here is that the exactness of (2) converts it from a phenomenon into a structural consequence, and that he draws the correct operational conclusion from it.

### 3.3 Thresholds and stability (§4)

Definition 4.1 introduces $g_{\mathrm{first}}(\delta t;\varepsilon)$ and $g_{\mathrm{term}}(\delta t;\varepsilon)$; the distinction is logically justified by Finding 1. Lemma 4.2 establishes Lipschitz dependence on $g$ with constant $D_{\mathrm{CK}} + M_*$, and the uniform pointwise estimate passes correctly to the supremum over the resolution set. Theorem 4.3 supplies the decreasing envelope $B(g) = (1-g)\big[D_{\mathrm{CK}}(\delta t;\Gamma_0) + M_*(\delta t)\big]$, the sufficient level $g_0 = 1 - \varepsilon/[D_{\mathrm{CK}}(\delta t;\Gamma_0) + M_*(\delta t)]$, and the chain

$$0 \;<\; g_{\mathrm{first}}(\delta t;\varepsilon) \;\leq\; g_{\mathrm{term}}(\delta t;\varepsilon) \;\leq\; 1 - \frac{\varepsilon}{D_{\mathrm{CK}}(\delta t;\Gamma_0) + M_*(\delta t)} \;<\; 1. \tag{4}$$

The strict-inequality issue at $g_0$ is handled correctly through the infimum definition.

Remark 4.5 is unusually candid about the weakness of its own bound. The estimate is *sign-blind*, so a larger interaction term makes the sufficient bound *weaker*, pushing the guarantee to stronger coupling rather than establishing suppression at smaller coupling; and the exact interaction term may align destructively with the substrate deficit and lower the actual thresholds substantially. In the paper's own interaction-dominant example the actual threshold is 0.108 against a bound of 0.834.

Proposition 4.9 (interval-local stability) shows that on a compact band of resolutions bounded away from zero, over which the unperturbed deficit remains uniformly above $\varepsilon$, no coupling weaker than an explicit $\eta(\varepsilon)$ can destroy detectability anywhere on the band. The use of Berge's maximum theorem on such intervals is valid, and the proposition follows from the reverse triangle inequality together with an explicit quadratic lower envelope. Remark 4.10 states plainly what the result does *not* say (no dynamical speed limit, no universal constant, no $\delta t \to 0$ asymptotics) and §7.3 confines the comparison with speed-limit theory to a single subsection and dismantles it there. This is a compact-interval stability result, not a dynamical speed limit, and the paper acknowledge this.

### 3.4 Causal lift and integrated deficit (§5)

The construction lifts to one-sided causal operators on $L^2([0,T]\times\Omega)$; the retarded operator is causal by construction and the perturbed retarded operator inherits causality. Theorem 5.3 correctly yields

$$J(\Gamma_g) \;\leq\; 2(1-g)^2 J(\Gamma_0) \;+\; 2g^2(1-g)^2 J_M,$$

whence $J(\Gamma_g) = O\big((1-g)^2\big)$. The paper is careful to state this as an upper rate rather than an exact one, noting that cancellation between the two terms of (2) can make the decay faster.

### 3.5 State-dependent and nonlinear coupling (§6)

Proposition 6.1 writes the state-dependent perturbation as a diagonal deformation, $\Gamma_{g(\cdot)}(t \leftarrow s) = \Gamma_0(I - G) + \Gamma_{\mathrm{ref}}G$, and derives a rigorous bound with explicit constants; the diagonal-column representation and midpoint comparison give the stated uniform remainder $R(\delta g) = O(\delta g)$. The result is correct. Section 6.2 handles nonlinear coupling through the generalized inverse $f^{\leftarrow}$ rather than $f^{-1}$, correctly observing that a continuous nondecreasing $f$ need not be injective; the transformation is valid for that class.

### 3.6 The discussion (§7): interpretation and its limits

Section 7 positions the results against the literature and is, in most respects, carefully done: the speed-limit comparison is explicitly disclaimed (§7.3); §7.1 declines to claim monotonicity of divisibility-based non-Markovianity functionals in $g$, citing the paper's own non-monotonicity; and §7.4 identifies the convex resource theory of Bhattacharya, Bhattacharya and Majumdar as "the closest precedent" and states three differences from it. Three items nevertheless require correction.

**Finding 2** (The inference "that is, at $g = 1$" is false, and the paper's own example refutes it)**.** The discussion states that exact divisibility, and with it information monotonicity, is restored where the deficit vanishes identically, "that is, at $g = 1$". The theory establishes divisibility *at* $g = 1$; it does not establish that $g = 1$ is the *only* such coupling, and no uniqueness argument is given. The inference is unjustified as stated.

It is also false on the paper's own data. In the interaction-dominant regime of §4.3 the exact deficit vanishes at the interior coupling $g = 0.3603$, which was verified independently by direct evaluation of the closed form; the paper itself reports the interior zero. A family whose deficit vanishes at an interior coupling is divisible at that coupling. The correct statement is: "in particular at $g = 1$, and at any other coupling for which the perturbed family is divisible."

The remedy is one clause, and no other result depends on the erroneous version.

**Finding 3** (The convex path is a formal perturbation, not a demonstrated realization)**.** Equation (1) always defines a valid two-time column-stochastic family. But a valid two-time family need not be the transition family of a single stochastic process with a compatible hierarchy of finite-dimensional distributions, and the paper supplies no realization theorem for $\Gamma_g$. This matters because §1 lists as the *first* of four contributions the formalization of the convex perturbation as "the canonical model of a dynamically consistent state-conditioned interaction".

The obstruction is concrete rather than pedantic. A global random choice between two dynamical branches, made once at the initial time, does not generically reproduce (1) with a *time-independent* weight: conditioning at a later time generally changes the posterior branch weight, so the induced two-time conditional kernels carry state- and time-dependent mixing weights rather than the same $g$ at every pair. Nor is the constant-$g$ form generically implied by measurement, control or state feedback.

The mathematical perturbation theory is unaffected — it is valid exactly as stated for the family (1), whatever that family's provenance. What must change is the claim attached to it. The author should either supply sufficient realization conditions, or state plainly that (1) is a formal perturbation path and drop "canonical". I note that §6 already uses the weaker and more defensible phrase "the leading-order model", which suggests the author's own view is closer to the correct one than §1 indicates.

**Finding 4** (An unresolved inconsistency about the meaning of the coupling)**.** The abstract and Definition 2.6 both insist that "observer coupling" is "a label only", carrying "no interpretive

weight in any statement or proof". Section 7.5 then draws precisely the interpretive conclusion: that detectability "depends on the perturbation budget available to the observer", and that "the perturbation modifies the family, not the apparatus". Both cannot stand. Given Finding 3, the disclaimer is the defensible half: absent an explicit implementation model, $g$ is best read as a mathematical mixture coordinate, and the §7.5 language should be withdrawn or made explicitly conditional on such a model.

**Scope of the no-generator statement.** The assertion that an indivisible substrate has "no generator reconstructing the two-time family" is too broad unless the generator class and the state space are specified. CK failure rules out an ordinary time-local evolution-family representation in the relevant class; it does not rule out memory kernels, Volterra descriptions, distributional generators, or augmented-state representations. The qualification costs one clause.

**Reproducibility and testability.** The algebra and the worked example are fully reproducible, and I reproduced them. The quantities $D_{\mathrm{CK}}$ and the coupling thresholds are in principle estimable from transition data. However, the paper supplies no finite-sample uncertainty analysis and no calibration of $\varepsilon$. Detectability as defined is therefore a deterministic, population-level notion, and becomes an empirical one only when supplemented by an external statistical procedure. The paper does not claim otherwise, but it should say so.

## 4 Originality: what is new and what is standard

**New.** The exact deficit identity (2) in the classical two-time column-stochastic setting, with its asymmetric factorization of the interaction term as a product of two substrate–reference differences — an asymmetry that reflects the divisibility of one summand and has no counterpart in a symmetric mixture of two general families. The observation of Finding 1 and its consequence for the formulation of detectability thresholds. The fixed-resolution $\varepsilon$-threshold formalism, and the separation of first-crossing from terminal thresholds, for which I found no direct antecedent. Proposition 6.1's explicit state-dependent bound.

**Standard, or adaptation of standard technique.** The suppression bound (3) is the triangle inequality applied to (2). Theorem 4.3 is a linear-envelope argument. Proposition 4.9 is, as the paper itself states, "a compactness-based stability statement" (that a continuous positive function on a compact set attains a positive minimum) with Berge's theorem supplying joint continuity. Remark 4.6's mixing estimates import Mitrophanov and Fill directly and are not used in the main theorem. The general idea of suppressing memory effects by mixing with Markovian dynamics is prior art (next section).

## 5 Priority in the literature

| **Element** | **Prior record** | **Assessment** |
|---|---|---|
| Convex combinations of dynamical maps; suppression and creation of non-Markovianity | Wudarski & Chruściński, *Phys. Rev. A* **93**, 042120 (2016); Jagadish, Srikanth & Petruccione, *Phys. Rev. A* **101**, 062304 (2020); Siudzińska, *Phys. Rev. A* **103**, 022605 (2021); Pathak & Shaji (2019) | Cited, and correctly characterised as *quantum CPTP* results. The classical column-stochastic, continuous two-time setting is genuinely different. Neither convex mixing nor non-monotonicity under mixing can be claimed as new |
| Convex resource theory of non-Markovianity | Bhattacharya, Bhattacharya & Majumdar, *J. Phys. A* **54**, 035302 (2021) | Cited as "the closest precedent", with three differences stated. The characterisation is accurate as far as it goes — but see the next row, which is closer still |
| =Robustness measures defined by the minimum Markovian admixture | N. Anand & T. A. Brun, "Quantifying non-Markovianity: a quantum resource-theoretic approach", arXiv:1903.03880 (2019), introducing *the robustness of non-Markovianity*, which "quantifies the minimum amount of Markovian noise that can be mixed with a non-Markovian evolution before it becomes Markovian". D. De Santis & V. Giovannetti, "Measuring non-Markovianity via incoherent mixing with Markovian dynamics", arXiv:2004.03741 (2020), *Phys. Rev. A* **103**, 012218 (2021), based on "the minimal amount of extra Markovian noise we have to add to the process via incoherent mixing, in order to make the resulting transformation Markovian too at all times" | Neither cited. Both read at source. De Santis–Giovannetti is the closest: a convex mixture with a *constant* weight, minimised subject to Markovianity *at all times*. This is conceptually the terminal-suppression question. **[R]**, and the most consequential item in this audit — see Finding 5 |
| Divisibility-based non-Markovianity functionals | Wolf et al.; Rivas, Huelga & Plenio; Chruściński & Maniscalco; Buscemi & Datta; Vacchini et al.; Benatti, Chruściński & Nichele (2024) | Cited. §7.1 correctly declines to claim monotonicity of such functionals in $g$, citing the paper's own non-monotonicity result |
| Cone-geometric characterisation of divisibility for two-state stochastic-quantum dynamics | Pimenta, arXiv:2505.08785 (2025) | Cited; I verified independently that this work is genuinely distinct (geometric and conical rather than operator-theoretic), and the paper flags the connection as an open question rather than claiming it |
| Product-form speed-limit inequalities | Mandelstam & Tamm; Margolus & Levitin; Shiraishi, Funo & Saito; Shanahan et al.; Wu & Yu; Nishiyama & Hasegawa | Cited *and explicitly disclaimed* in §7.3. Correct handling; no priority issue arises |

| Element | Prior record | Assessment |
|---|---|---|
| Contemporaneous neighbouring work | W. Xu & V. Jagadish, "Convexity and non-Markovianity of Weyl Maps", arXiv:2605.23852 (22 May 2026): convex combinations of eternally non-Markovian Weyl dephasing maps can generate Markovian semigroups, so "non-Markovianity is not additive under mixing" | Not cited. It reproduces neither the CK identity nor the threshold theorem, but it further limits any broad claim about convex suppression of memory effects. **[R]** (framing and positioning) |

**Finding 5** (The terminal-suppression concept is not prior-art-free)**.** It is tempting to conclude that no priority question arises here, on the ground that the prior art on convex mixing is uniformly quantum and CPTP while the classical two-time column-stochastic formulation is the author's own. That reasoning does not survive scrutiny. The relevant question is not whether the *setting* is new but whether the *concept* of a minimal Markovian admixture that removes memory is new, and it is not. Anand and Brun introduced the robustness of non-Markovianity as exactly the minimum Markovian noise that renders an evolution Markovian; De Santis and Giovannetti pose the constant-weight version and minimise it subject to Markovianity at all times. The manuscript's terminal-suppression concept is therefore not prior-art-free.

The omission is the more conspicuous because §7.4 already describes the terminal-threshold theorem as "in the same family as the robustness-of-non-Markovianity result" of Bhattacharya et al. ( that is, the author has correctly identified the family of results while omitting the two papers in which the measure originates and takes its closest form).

**What survives.** The defensible novelty lies in fixing a specific divisible reference; in working with a classical two-time CK operator-norm defect rather than CP-divisibility of quantum maps; in allowing $\varepsilon$-level suppression rather than exact Markovianization, which is what makes the two-threshold structure meaningful; and in resolving the criterion at a chosen temporal resolution $\delta t$. These distinctions are real and support moderate originality. They must be stated *against* the missing references rather than in their absence.

## 6 Specific criticisms

Items are marked **[M]** where they bear on the correctness or the substance of a claim, and **[R]** where they concern presentation, attribution or scope.

**[M] The claim that the convex mixture is "the canonical model" of a dynamically consistent state-conditioned interaction is not established.**
No realization theorem for $\Gamma_g$ is supplied, and a global branch choice does not generically produce a time-independent weight (Finding 3). Remedied either by sufficient realization conditions or by recasting (1) as a formal perturbation path.

**[M] The inference "that is, at $g = 1$" is false**
and is contradicted by the paper's own §4.3 example, in which the exact deficit vanishes at $g = 0.3603$ (Finding 2). This is the least serious of the material items — the remedy is a single clause — but it is a false statement rather than an imprecision.

**[R] Closest prior art on minimal Markovian admixture not cited.**
Anand & Brun (2019) and De Santis & Giovannetti (2020/2021) (Finding 5). The most consequential of the revisable items.

**[R] Unresolved inconsistency about the meaning of the coupling**
(Finding 4).

**[R] The no-generator statement is unqualified.**
Specify the generator class and state space; CK failure excludes an ordinary time-local evolution-family representation, not memory kernels, Volterra descriptions, distributional generators or augmented-state representations.

**[R] No finite-sample calibration of $\varepsilon$.**
Detectability as defined is a deterministic population-level notion; the paper should say so, or supply the statistical procedure.

**[R] Xu & Jagadish (arXiv:2605.23852) not engaged.**

**[R] Modest mathematical depth.**
Correct throughout, but the content is an exact two-line expansion, its triangle inequality, a linear envelope and a compactness argument. This bears on weight, not on correctness.

# 7 Questions to the author

1. Give a stochastic-process realization producing the same constant $g$ in every two-time conditional kernel, or explain why none is required and recast (1) as a formal perturbation path. State explicitly whether the families are assumed process-induced, and what remains valid if they are abstract two-time stochastic families instead.

2. Explain the mathematical distinction between your terminal threshold and the robustness measures of Anand & Brun and of De Santis & Giovannetti, and add both references. The distinctions I consider defensible are set out in Finding 5; confirm or replace them.

3. Correct the assertion that exact restoration occurs "that is, at $g = 1$". Can $\Gamma_g$ become exactly divisible at an interior coupling? Your own §4.3 example says it can; state what, if anything, characterises the set of such couplings.

4. Reconcile the disclaimer that "observer coupling" carries no interpretive weight with the interpretive conclusions drawn in §7.5.

5. Qualify the no-generator statement: which generator classes are actually excluded by CK failure?

6. How would $\varepsilon$ be calibrated from finite-sample transition estimates, and what null hypothesis would be tested?

7. What invariant meaning, if any, should be attached to the reference-dependent interaction scale $M_*$? Can the terminal bound be sharpened using operator direction rather than a sign-blind norm bound? Remark 4.5 reports a factor-of-eight gap in your own example, which suggests there is room.

8. Keep the speed-limit comparison strictly analogical, as §7.3 already does, and state whether you intend to pursue the rate-form question or to withdraw it.

# 8 Technical conclusion

The central results are mathematically sound. The exact deficit identity is correct and, in the classical two-time column-stochastic form given here, appears to be the author's own; the derivation of the two-threshold structure from the exactness of that identity is a genuine methodological point, correctly argued. The paper is disciplined about its own limitations in several places where a less careful author would have claimed more (the sign-blindness of the principal bound, the disclaimer of speed-limit content, the explicit imposition rather than derivation of dynamical consistency for the reference).

Two matters require substantive correction: the status of the convex mixture as a model of a physical interaction, which is asserted rather than proved, and a false uniqueness inference in the discussion which the paper's own worked example refutes. A third, the priority positioning against the minimal-Markovian-admixture literature, requires the addition of two references and an explicit statement of what remains distinctive. None of these requires new mathematics, and none disturbs the identity, the bounds, or the threshold theorem.